

New Moving Cloud Reasoning Method and Stabilization Control of the Inverted Pendulum

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Abstract

In this paper, we propose a new moving cloud reasoning method that can show a novel research direction in the approximate inference of nonlinear systems with uncertainty. In 1995, Professor De-Yi Li suggested the cloud model as an uncertainty conversion model between the qualitative and the quantitative—that is, the membership cloud model—to represent and process uncertainty by fuzziness and randomness on the basis of probability statistics and the traditional fuzzy set theory suggested by Zadeh. However, the membership cloud model has the weakness that a large amount of derivative and integral computation must be carried out to obtain the entropy of the membership cloud, and there are some errors in the condition expressions of the formulation of the preceding moving cloud reasoning method. In this paper, as one approach to overcoming the weaknesses of the preceding papers, we propose a new approach of moving cloud reasoning to obtain the entropy of the membership cloud that does not use derivative and integral computation, which is based on the compensation fuzzy reasoning method (moving-rate principle) proposed by Son-Il Kwak et al. in 2019. The proposed m -input single-output seven-step reasoning algorithm reflects fuzziness and randomness as well as the three numerical characteristics of the cloud (expected value, entropy, and hyper entropy) through cloudification and de-cloudification of the input information, and satisfies the reductive property. Through experimental research on the stabilization control of a single inverted pendulum system, the effectiveness of the proposed approach is demonstrated: the range of the hyper entropy for which the proposed method converges is $[0.03, 0.50]$, considerably wider than the range $[0.04, 0.23]$ of the preceding moving cloud reasoning method; the average calculation time of 20 operations is 742 ms for the proposed method against 2058 ms for the preceding moving method and 1375 ms for T-S cloud modeling; and the overshoot of the pendulum and cart is substantially reduced.

Keywords: membership cloud; cloud model; moving cloud reasoning; entropy; hyper entropy; reductive property; inverted pendulum

1 Introduction

Since Zadeh suggested the concept of the fuzzy set in 1965 to process the uncertainty existing in nature and society [1], fuzzy theory has made rapid progress and found wide application during half a century [2, 13]. Fuzzy theory processes uncertainty with a fuzzy set; however, the decision of the membership function may look differently according to each person’s subjective point of view: after a membership function is decided, it becomes crisp. That is, fuzziness may be seen as a part of crisp mathematics, and this is a weakness of fuzzy theory.

In 1995, Professor De-Yi Li formulated an uncertainty conversion model between the qualitative and the quantitative to represent and process uncertainty by fuzziness and randomness

on the basis of probability and statistics: the *cloud model* [2, 15]. The cloud model is a cognitive model for qualitative–quantitative transformation, mathematically based on probability and statistics, that corresponds to qualitative concepts in human cognition. On this basis the virtual cloud, the integrated cloud, and the traditional cloud generators were developed, and the scientific level of uncertainty information processing was raised further [3]. Under the notion of the cloud model, uncertainty reasoning methods based on the cloud were defined and have been efficiently applied in various fields: intelligent control, human recognition and thought activities, image segmentation [4], granular computing [5], recursive integral proportional–integral control for active power filters [6], deep machine learning for passenger profiling [7], and the estimation of the reliability of computer systems [5, 13].

The inverted pendulum system is the classical benchmark of nonlinear, unstable, underactuated control. It reflects many important control problems, such as robustness, adaptation, and tracking [9]. Almost all control approaches—PID, LQ, adaptive, robust, H_∞ , neural network, fuzzy, GA, PSO, grey, chaos, and nonlinear control—have been applied to the stabilization control of the inverted pendulum system, and it is widely used in the development and validity checking of modern control theory; thus it has become one of the most important problems in the field of control science [10, 11].

However, it is difficult to apply cloud theory in practice because of the *entropy* peculiar to cloud theory compared with fuzzy theory: derivatives and integrals are used to obtain the certainty degree of each point of the membership cloud, which entails an enormous calculation amount. Moreover, there are some errors in the condition expressions of the formulation of the preceding moving cloud reasoning method. These weaknesses of the preceding papers must be eliminated. In this paper, we eliminate the errors and propose an efficient moving cloud reasoning approach, and we verify its effectiveness by applying it to the stabilization control of a single inverted pendulum system. The approach is based on the compensation fuzzy reasoning principle (moving rate) proposed by Son-Il Kwak et al. in 2019 [12].

The main contributions of this paper are: (i) a derivative- and integral-free computation of the entropy of the membership cloud, obtained by the moving principle, which drastically reduces the computational burden of cloud reasoning; (ii) a corrected formulation of the condition expressions of the preceding moving cloud reasoning method; (iii) a complete seven-step m -input single-output moving cloud reasoning algorithm that preserves the three numerical characteristics of the cloud (expected value E_x , entropy E_n , hyper entropy H_e) and satisfies the reductive property; and (iv) an experimental verification on the stabilization control of the single inverted pendulum, showing a wider convergent hyper-entropy range, smaller overshoot, and an almost threefold reduction of computation time with respect to the preceding moving method.

The rest of the paper is organized as follows. Section 2 reviews related work on cloud models, uncertain fuzzy reasoning, and inverted pendulum control, and states the research gap. Section 3 presents the novel moving cloud reasoning method. Section 4 designs the stabilization controller of the single inverted pendulum system based on the moving cloud reasoning method. Section 5 reports the results and discussion, and Section 6 concludes the paper.

2 Related Work and Motivation

2.1 Cloud model theory and its applications

The cloud model introduced by Li [2, 15] describes a qualitative concept by three numerical characteristics—expected value E_x , entropy E_n , and hyper entropy H_e —and generates cloud

drops by a two-stage normal random process, thereby unifying fuzziness and randomness in a single framework. Zhou studied the cloud model and its application in intelligent control [3], and Liu investigated granular computing based on the cloud [5]. Application studies have flourished: Li *et al.* synergistically integrated graph-cut and cloud model strategies for image segmentation [4]; Li and Fan realized recursive integral proportional–integral control based on the membership cloud for active power filters [6]; and, more recently, Jia *et al.* proposed a situation assessment method for aerial targets based on linguistic fuzzy sets and trapezium clouds [23], confirming that cloud-based uncertainty representation remains an active research direction. Nevertheless, in all of these works the certainty degree and the entropy of the membership cloud are evaluated through derivative and integral computation, whose cost grows quickly with the number of cloud drops and rules; this is the principal obstacle to real-time cloud reasoning in control loops.

2.2 Fuzzy reasoning and identification under uncertainty

Approximate reasoning has been studied intensively since Zadeh’s compositional rule of inference [1], and the classical comparison of fuzzy reasoning methods was given by Mizumoto and Zimmermann [13]. Modern developments include recursive fuzzy instrumental-variable-based evolving neuro-fuzzy identification for non-stationary dynamic systems in noisy environments [8], ANFIS-based model predictive control for chemical reactors [9], risk-attitudinal Pythagorean fuzzy decision models [10], and interval type-2 fuzzy tracking control via sampled-data controllers [11]. T–S fuzzy model identification with a gravitational-search-based hyperplane clustering algorithm was proposed in [14] and is used as a comparison method in this paper.

In parallel, our group has established the *moving-rate (compensating-operation) reasoning principle*: a fuzzy reasoning method based on the compensating operation was proposed and applied to fuzzy systems in [12]; a distance-measure-based reasoning method with a proven reductive property was presented in [16]; fuzzy modus ponens and modus tollens based on moving distance in the SISO fuzzy system were established in [17]; an extended-distance-measure reasoning method for discrete fuzzy vectors of different dimensions was reported in [18]; and a least-common-multiple-based reasoning method with its property analysis was given in [19]. The principle has recently been extended to interval fuzzy modeling with application to turf color image evaluation [20], and related work of our group on explainable and optimally tuned fuzzy systems includes a causal-entropic Bayesian framework for robust fuzzy inference [21] and improved shark smell optimization with adaptive parameter scheduling for fuzzy controller tuning [22]. The present paper transplants this moving principle from fuzzy sets to membership clouds.

2.3 Stabilization control of the inverted pendulum

As the canonical validation platform of nonlinear control, the inverted pendulum continues to attract new methods. Su *et al.* designed event-triggered fuzzy control of nonlinear systems with application to inverted pendulum systems [24]; Nasir and Razak optimized a type-2 fuzzy controller for the inverted pendulum with an opposition-based spiral dynamic algorithm [25]; Safeea and Neto applied Q-learning to the continuous control problem of robot inverted pendulum balancing [26]; Irfan *et al.* carried out a comparative analysis of linear, nonlinear, and artificial-intelligence control strategies for the inverted pendulum [27]; Nguyen *et al.* combined optimized fuzzy logic and sliding-mode control for stability and disturbance rejection in the rotary inverted pendulum [28]; Saleem *et al.* validated an EKF-based fuzzy adaptive sliding-mode controller experimentally [29]; and Tian *et al.* developed an RBF-ARX model-based predictive

control approach with a self-triggered mechanism [30]. These recent studies show that interpretable rule-based controllers with modest computational cost remain highly competitive on this benchmark, which motivates the use of the inverted pendulum for validating the proposed cloud reasoning method.

2.4 Research gap and contributions

The literature review reveals the following gap. Cloud-model reasoning [2, 3, 4, 5, 6, 15, 23] represents both fuzziness and randomness but pays for this with derivative/integral-based entropy computation that is too heavy for real-time control; classical and modern fuzzy reasoning [1, 8, 9, 10, 11, 13, 14] is computationally light but does not carry the randomness (hyper-entropy) information of the cloud; and the preceding moving cloud reasoning method contains errors in its condition expressions and converges only on a narrow hyper-entropy range. The moving-rate principle [12, 16, 17, 18, 19, 20] has so far been developed only for fuzzy sets. This paper closes the gap by carrying the moving-rate principle over to membership clouds, yielding a moving cloud reasoning method that is derivative- and integral-free, correctly formulated, convergent on a wide hyper-entropy range, and fast enough for real-time stabilization control.

3 A Novel Moving Cloud Reasoning Method

In this paper, unlike the papers [3, 5, 6, 10, 15], we propose a new moving cloud reasoning method. Let us represent the simplest m -input, single-output, single-rule cloud reasoning model by expression (1):

$$\text{If } x_1 \text{ is } A_1, \dots, x_m \text{ is } A_m, \text{ then } y \text{ is } B. \quad (1)$$

The new cloud reasoning method is as follows.

Step 1. For expression (1), obtain the expected curves of the m premise membership clouds A_j , $j = \overline{1, m}$, and of the consequence membership cloud B according to expressions (2) and (3), where x_{j0} is the expected value of the premise membership cloud A_j and y_0 is the expected value of the membership cloud B :

$$\mu_{A_j}(x) = \exp \left[\frac{-(x - x_{j0})^2}{2b_j^2} \right], \quad (2)$$

$$\mu_B(y) = \exp \left[\frac{-(y - y_0)^2}{2b^2} \right]. \quad (3)$$

Step 2. Let $A_j E_n(x)$ be the entropies of the m premise membership clouds A_j ; obtain them according to expression (4):

$$A_j E_n(x) = \sigma_{A_j, \max} \exp \left[\frac{-(\Delta L_{A_j}(x))^2}{2M_{A_j}^2(x)} \right], \quad (4)$$

where $\sigma_{A_j, \max}$ is the hyper entropy of the membership cloud A_j . In expression (4), $\Delta L_{A_j}(x)$ and $M_{A_j}(x)$ are decided in the universal domain of the membership cloud A_j as follows:

$$\Delta L_{A_j}(x) = r_j(x) \Delta L_{A_j}^l(x) + [1 - r_j(x)] \Delta L_{A_j}^r(x), \quad (5)$$

$$\Delta L_{A_j}^r(x) = x - x_{j0} + \sqrt{\ln 8} b_j, \quad (6)$$

$$M_{A_j}(x) = r_j(x) M_{A_j}^l(x) + [1 - r_j(x)] M_{A_j}^r(x), \quad (7)$$

$$r_j(x) = \min[1 - \text{sign}(x - x_{j0}), 1]. \quad (8)$$

This expression, as a term to decide left hand or right hand according to the sign of $\text{sign}(x - x_{j0})$, has 0 or 1 value. In expression (7), let $M_{A_j}^l(x)$ be the left width of the membership cloud centering on the point x_{j0} on the axis and $M_{A_j}^r(x)$ be the right width of the membership cloud A_j ; then the following relation holds:

$$M_{A_j}^l(x) = M_{A_j}^r(2x_{j0} - x). \quad (9)$$

The right width $M_{A_j}^r(x)$ of the membership cloud A_j is obtained as follows:

$$M_{A_j}^r(x) = p_j(x) \sqrt{\ln 8} b_j + [1 - p_j(x)](x - x_{j0}), \quad (10)$$

$$p_j(x) = \min[1 - \text{sign}(x - x_{j0} - \sqrt{\ln 8} b_j), 1]. \quad (11)$$

Step 3. Obtain the entropy $BE_n(y)$ of the membership cloud B according to expression (12):

$$BE_n(y) = \sigma_{B,\max} \exp \left[\frac{-(\Delta L_B(y))^2}{2 M_B^2(y)} \right], \quad (12)$$

where $\sigma_{B,\max}$ is the hyper entropy of the membership cloud B , and

$$\Delta L_B(y) = r(y) \Delta L_B^l(y) + [1 - r(y)] \Delta L_B^r(y), \quad (13)$$

$$\Delta L_B^r(y) = y - y_0 + \sqrt{\ln 8} b. \quad (14)$$

In expression (12), $M_B(y)$ is the width decided in the universal domain of the membership cloud B :

$$M_B(y) = r(y) M_B^l(y) + [1 - r(y)] M_B^r(y), \quad (15)$$

where y_0 is the center of the membership cloud B and $r(y)$, which has the same meaning as expression (8), is

$$r(y) = \min[1 - \text{sign}(y - y_0), 1]. \quad (16)$$

Let $M_B^l(y)$ be the left width of the membership cloud B centering on the point y_0 of the horizontal axis and $M_B^r(y)$ be the right width of B ; then the following relation holds:

$$M_B^l(y) = M_B^r(2y_0 - y). \quad (17)$$

The right width $M_B^r(y)$ of the membership cloud B is obtained as follows:

$$M_B^r(y) = p(y) \sqrt{\ln 8} b + [1 - p(y)](y - y_0), \quad (18)$$

$$p(y) = \min[1 - \text{sign}(y - y_0 - \sqrt{\ln 8} b), 1]. \quad (19)$$

Step 4. Obtain the premise membership clouds $A'_1, \dots, A'_m$, which cloudify the crisp input information $x_1^0, \dots, x_m^0$, and their entropies $A'_j E_n(x)$ as shown above:

$$\mu_{A'_j}(x) = \exp \left[\frac{-(x - x_j^0)^2}{2 (b'_j)^2} \right], \quad (20)$$

where the input information $x_1^0, \dots, x_m^0$ has the meaning of the expected values of the premise

membership clouds $A'_1, \dots, A'_m$. The entropy of the premise membership cloud A'_j which cloudifies the input information is as follows:

$$A'_j E_n(x) = \sigma_{A'_j, \max} \exp \left[\frac{-(\Delta L_{A'_j}(x))^2}{2 M_{A'_j}^2(x)} \right], \quad (21)$$

where $\sigma_{A'_j, \max}$ is the hyper entropy of the membership cloud A'_j . In expression (21), $\Delta L_{A'_j}(x)$ and $M_{A'_j}(x)$ are as follows:

$$\Delta L_{A'_j}(x) = r'_j(x) \Delta L_{A'_j}^l(x) + [1 - r'_j(x)] \Delta L_{A'_j}^r(x), \quad (22)$$

$$\Delta L_{A'_j}^r(x) = x - x_j^0 + \sqrt{\ln 8} b'_j, \quad (23)$$

$$M_{A'_j}(x) = r'_j(x) M_{A'_j}^l(x) + [1 - r'_j(x)] M_{A'_j}^r(x), \quad (24)$$

$$r'_j(x) = \min[1 - \text{sign}(x - x_j^0), 1]. \quad (25)$$

In expression (24), let $M_{A'_j}^l(x)$ be the left width of the membership cloud centering on the point x_j^0 on the axis and $M_{A'_j}^r(x)$ be the right width of the membership cloud A'_j ; then the following relation holds:

$$M_{A'_j}^l(x) = M_{A'_j}^r(2x_j^0 - x). \quad (26)$$

Obtain the right width $M_{A'_j}^r(x)$ of the membership cloud A'_j as follows:

$$M_{A'_j}^r(x) = p'_j(x) \sqrt{\ln 8} b'_j + [1 - p'_j(x)](x - x_j^0), \quad (27)$$

$$p'_j(x) = \min[1 - \text{sign}(x - x_j^0 - \sqrt{\ln 8} b'_j), 1]. \quad (28)$$

Step 5. Obtain the normal random numbers $A_j N R N_k$ and the cloud moving function $A_j C M F(x_k)$ according to expressions (29) and (30) for the cloud moving method:

$$A_j N R N_k = J[A_j E_n(x_k), A'_j E_n(x_k), A_j H_e(x), A'_j H_e(x)], \quad k = \overline{1, g}. \quad (29)$$

In expression (29), $J[\cdot]$ means the normal random numbers obtained by $A_j E_n(x_k)$, $A'_j E_n(x_k)$, $A_j H_e(x)$, and $A'_j H_e(x)$, where $A_j E_n(x)$ is the entropy of the premise membership cloud A_j , $A'_j E_n(x)$ is the entropy of the membership cloud A'_j of the input information, $A_j H_e(x)$ is the hyper entropy of the premise membership cloud A_j , and $A'_j H_e(x)$ is the hyper entropy of the membership cloud A'_j of the input information. Obtain the membership cloud moving function $A_j C M F(x_k)$ as follows:

$$A_j C M F(x_k) = [x_j^0 - A_j E_x] \cdot A_j N R N_k, \quad k = \overline{1, g}, \quad (30)$$

where $A_j E_x = x_{j0}$ is the expected value of the premise membership cloud A_j .

Step 6. On the basis of expression (30), compute the cloud moving actions of the conclusion part of the rule, $A_j C M A$, as expression (31), and obtain the total cloud moving action $A C M A(y)$ of the conclusion part in terms of the arithmetic average of the moving actions:

$$A_j C M A = \frac{1}{g} \sum_{k=1}^g A_j C M F(x_k), \quad (31)$$

$$ACMA(y) = \frac{1}{m} \sum_{j=1}^m A_j CMA. \quad (32)$$

The membership cloud B' of the reasoning result is generated by moving the cloud B by $ACMA(y)$:

$$\mu_{B'}(y) = \mu_B(y - ACMA(y)). \quad (33)$$

Obtain the entropy $B'E_n(y)$ of the membership cloud B' of the reasoning result according to expression (34):

$$B'E_n(y) = \sigma_{B', \max} \exp \left[\frac{-(\Delta L_{B'}(y))^2}{2 M_{B'}^2(y)} \right], \quad (34)$$

where $\sigma_{B', \max}$ is the hyper entropy of the membership cloud B' obtained by the reasoning result, and

$$\Delta L_{B'}(y) = r(y) \Delta L_{B'}^l(y) + [1 - r(y)] \Delta L_{B'}^r(y), \quad (35)$$

$$\Delta L_{B'}^r(y) = y - y' + \sqrt{\ln 8} b. \quad (36)$$

In expression (34), $M_{B'}(y)$ is the width decided in the universal domain of the membership cloud B' :

$$M_{B'}(y) = r(y) M_{B'}^l(y) + [1 - r(y)] M_{B'}^r(y), \quad (37)$$

where y' is the center of the membership cloud B' and, in expression (37), $r(y)$ is represented by

$$r(y) = \min[1 - \text{sign}(y - y'), 1]. \quad (38)$$

Let $M_{B'}^l(y)$ be the left width of the membership cloud B' centering on the point y' of the horizontal axis and $M_{B'}^r(y)$ be the right width of the membership cloud B' ; then

$$M_{B'}^l(y) = M_{B'}^r(2y' - y). \quad (39)$$

Obtain the right width $M_{B'}^r(y)$ of the membership cloud B' of the reasoning result according to expression (40):

$$M_{B'}^r(y) = p(y) \sqrt{\ln 8} b + [1 - p(y)](y - y'), \quad (40)$$

$$p(y) = \min[1 - \text{sign}(y - y' - \sqrt{\ln 8} b), 1]. \quad (41)$$

Step 7. Obtain the crisp output y' , which de-cloudifies the moving cloud reasoning result B' , as expression (42):

$$y' = y_0 + ACMA(y). \quad (42)$$

Note that all quantities in Steps 2–7 are obtained from the expected curves, the sign functions, and arithmetic averages only; no derivative or integral computation of the certainty degree appears anywhere in the algorithm. This is the essential difference from the preceding cloud reasoning methods and the source of the computational savings reported in Section 5.

4 Stabilization Control of a Single Inverted Pendulum System Based on the Moving Cloud Reasoning Method

4.1 Working principle of stabilization

The geometric configuration diagram of the single inverted pendulum is shown in Figure 1, and the inverted pendulums in different initial states are shown in Figure 2.

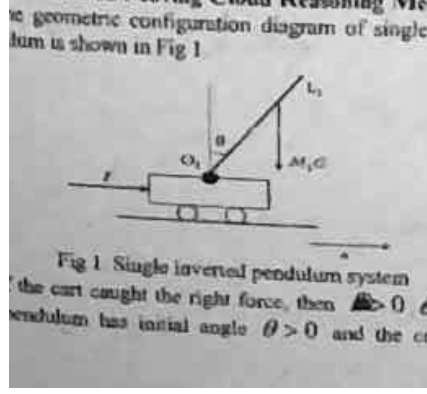


Figure 1: Single inverted pendulum system.

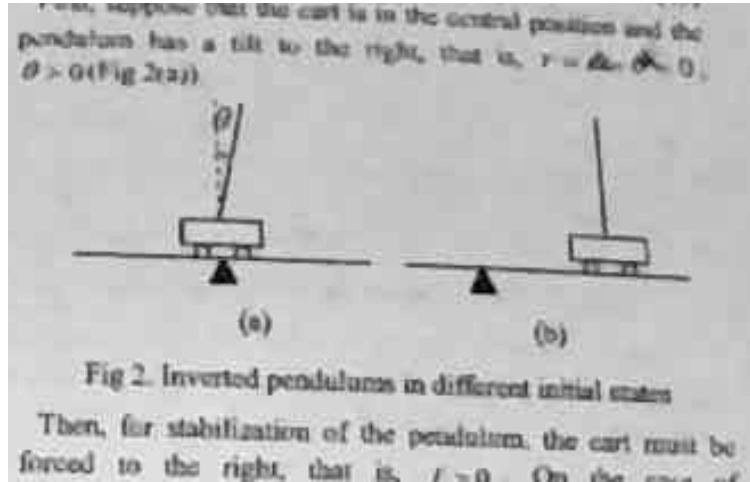


Figure 2: Inverted pendulums in different initial states.

If the cart catches the right force, the pendulum has initial angle $\theta > 0$; if the cart never catches any force, the pendulum falls down in terms of gravity. These relations may be represented by the following expression:

$$\ddot{\theta} = \delta_1 \dot{\theta} + \delta_2 \theta, \quad \delta_1 > 0, \delta_2 > 0. \quad (43)$$

First, suppose that the cart is in the central position and the pendulum has a tilt to the right, that is, $r = \dot{r} = \dot{\theta} = 0, \theta > 0$ (Fig. 2(a)). Then, for stabilization of the pendulum, the cart must be forced to the right, that is, $f > 0$. In the case of $r = \dot{r} = \dot{\theta} = 0, \theta < 0$, for stabilization of the pendulum the cart must catch the force $f < 0$ as well. That is,

$$f \sim k_1 \theta + k_2 \dot{\theta}. \quad (44)$$

Suppose now that the pendulum is in the stable state and the cart is off the center, that is, $\theta = \dot{\theta} = \dot{r} = 0$, $r > 0$ (Fig. 2(b)). In this case, for stabilization of the cart, it must be forced to the left; however, θ then gets large in terms of expression (43). Since stabilization of the pendulum is the first purpose, by expression (44) the cart catches the force $f > 0$ and the cart moves right; in consequence, the position of the cart is not stabilized. Hence, the cart must first be forced to the right. The case of $\dot{r} < 0$ is interpreted like the preceding. That is,

$$f = k_1\theta + k_2\dot{\theta} + k_3r + k_4\dot{r}, \quad (k_1, k_2, k_3, k_4 > 0). \quad (45)$$

4.2 Design of the stabilization controller by the moving cloud reasoning method

The mathematical model of the single inverted pendulum is represented as follows:

$$k_1\ddot{q} + k_2\dot{q} + k_3(q, \dot{q}) = k_4u, \quad k_1 = \begin{bmatrix} m_1 + m_2 & -m_2l \cos \theta \\ -m_2l \cos \theta & \frac{4}{3}m_2l^2 \end{bmatrix}, \quad q = [r, \theta]^T, \quad (46)$$

where r is the position of the cart, θ is the angle of the pendulum, m_1 is the mass of the cart, m_2 is the mass of the pendulum, l is the length of the pendulum, and $g = 9.8 \text{ m/s}^2$. Then the nonlinear state equation of the single inverted pendulum system is represented as follows:

$$\dot{x}(t) = (A(x) + \Delta A)x(t) + (B(x) + \Delta B)u(t), \quad (47)$$

where $A(x) = \begin{bmatrix} 0 & I \\ -k_1^{-1}k_3 & -k_1^{-1}k_2 \end{bmatrix}$ and $B(x) = \begin{bmatrix} 0 \\ k_1^{-1}k_4 \end{bmatrix}$.

A nonlinear model of the object is approximately represented by the following three rules:

$$\begin{aligned} R^1 : & \text{ if } \theta \text{ is } N \text{ then } \dot{x}(t) = (A_1 + \Delta A_1)x(t) + (B_1 + \Delta B_1)u(t), \\ R^2 : & \text{ if } \theta \text{ is } Z \text{ then } \dot{x}(t) = (A_2 + \Delta A_2)x(t) + (B_2 + \Delta B_2)u(t), \\ R^3 : & \text{ if } \theta \text{ is } P \text{ then } \dot{x}(t) = (A_3 + \Delta A_3)x(t) + (B_3 + \Delta B_3)u(t), \end{aligned} \quad (48)$$

where N , Z , and P are the membership clouds of the premise variable θ shown in Figure 3. The control amount by the cloud model controller is obtained by the proposed moving method of cloud reasoning; in other words, the clouds are generated differently from the preceding methods, and the membership cloud of the conclusion part is generated by cloud moving.

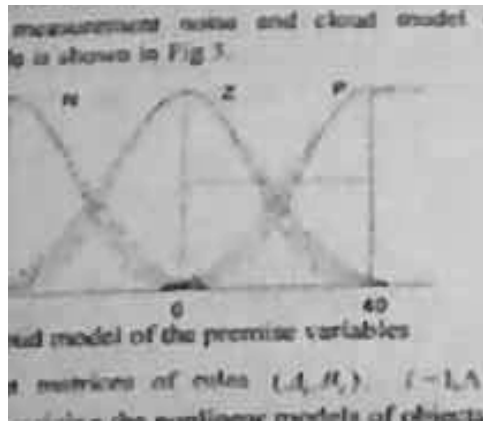


Figure 3: Cloud model of the premise variables.

The coefficient matrices of the rules (A_i, B_i) , $i = \overline{1, 3}$, are obtained by linearizing the non-linear model of the object in the neighborhoods of $x_1 = (0, \pm\pi/6, 0)^T$ and $x_2 = (0, 0, 0)^T$, respectively:

$$A_1 = A_3 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0.861 & -7.929 & -0.0348 \\ 0 & 48.72 & -34.81 & -1.971 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0.9466 & -8.019 & -0.037 \\ 0 & 50.79 & -32.51 & -1.993 \end{bmatrix},$$

$$B_1 = B_3 = [0 \ 0 \ 0.793 \ 3.487]^T, \quad B_2 = [0 \ 0 \ 0.8 \ 3.775]^T, \quad C_i = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}.$$

Fix the control rules as follows:

$$\begin{aligned} L^1 : \text{if } \theta \text{ is } N \text{ then } \dot{\hat{x}}(t) &= A_{c1}\hat{x}(t) + B_{c1}y, \quad u(t) = C_{c1}\hat{x}(t), \\ L^2 : \text{if } \theta \text{ is } Z \text{ then } \dot{\hat{x}}(t) &= A_{c2}\hat{x}(t) + B_{c2}y, \quad u(t) = C_{c2}\hat{x}(t), \\ L^3 : \text{if } \theta \text{ is } P \text{ then } \dot{\hat{x}}(t) &= A_{c3}\hat{x}(t) + B_{c3}y, \quad u(t) = C_{c3}\hat{x}(t). \end{aligned}$$

Obtain $P > 0$, $Q > 0$ and the gains K_i of each control rule; then

$$\begin{aligned} A_{c1} = A_{c3} &= \begin{bmatrix} 0.8574 & -0.0101 & 0.0235 & -0.0133 \\ -0.0614 & 0.9619 & 0.0295 & -0.0145 \\ -0.6031 & -0.0946 & 0.8339 & 0.0926 \\ 0.7189 & -0.0318 & 0.1020 & 0.9232 \end{bmatrix}, \\ A_{c2} &= \begin{bmatrix} 0.8515 & -0.0315 & 0.0241 & -0.0134 \\ -0.0626 & 0.9627 & 0.0290 & -0.0140 \\ -0.6284 & -0.0945 & 0.8293 & 0.0945 \\ 0.7306 & -0.0289 & 0.1048 & 0.9221 \end{bmatrix}. \end{aligned}$$

5 Results and Discussion

We applied the proposed approach to the stabilization control of the inverted pendulum system for a performance comparison of the proposed moving cloud reasoning method with the preceding methods [11, 14]. Figures 4–9 show the results of the preceding method and the proposed method in the form of graphs for the cart and the pendulum (left: cart, right: pendulum).

The range of hyper entropy on which the preceding method can converge is $[0.04, 0.23]$, established through the experimental results of the preceding moving cloud reasoning method compared with the proposed approach, while the proposed moving cloud reasoning method converges on the range $[0.03, 0.50]$. The subsequent figures show the comparison result of T–S cloud modeling [14] and the proposed moving method of cloud reasoning (red: T–S cloud modeling [14]; black: proposed method). Table 1 shows the comparison result of the three methods (object 1: cart; object 2: pendulum; Ov: overshoot amount; Tr: transient time).

Regarding the calculation time, the average calculation time of 20 operations is 1375 ms for T–S cloud modeling, 2058 ms for the preceding moving cloud reasoning method, and 742 ms for the proposed moving cloud reasoning method. For the three methods mentioned above, the comparison of the overshoot amount and the transient time of the pendulum and the cart is shown in Table 1.

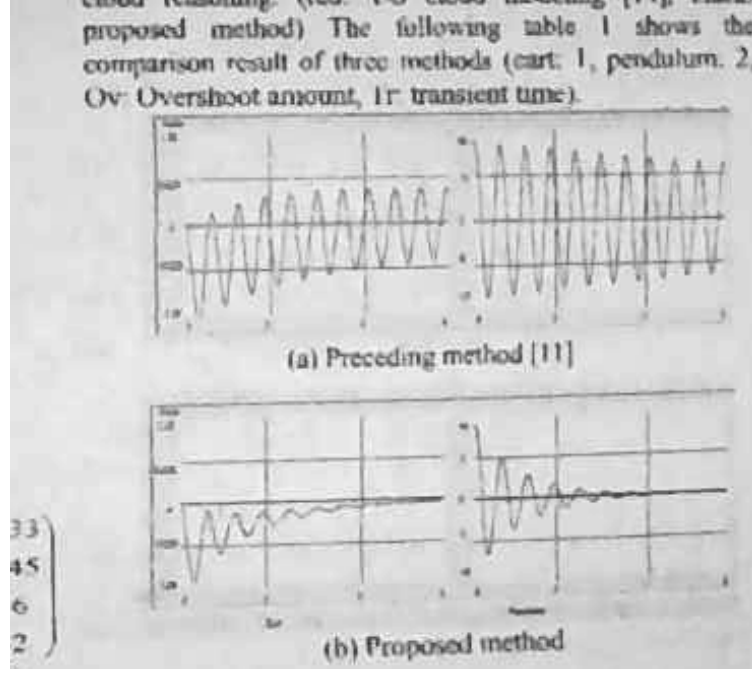


Figure 4: Result of the case that the hyper entropy is 0.04: (a) preceding method [11]; (b) proposed method.

Table 1: Comparison of the proposed method and the preceding method [14] (excerpt; Ov: overshoot, Tr: transient time, s).

Hyper entropy	Obj.	T-S cloud modeling [14]		Proposed method		Improvement	
		Ov	Tr	Ov	Tr	Ov %	Tr %
0.01	1	1.11	4.83	divergence of [14]			
	2	6.93	4.22	—			
0.02	1	1.11	4.83	divergence of [14]			
	2	6.93	4.27	—			
0.13	1	1.11	4.83	0.89	4.60	19.1	5
	2	6.93	4.22	0.83	1.65	43.1	10
0.14	1	1.11	4.83	0.83	4.60	11.2	13.2
	2	6.93	4.22	0.85	4.49	—	—

Three observations follow from these results. First, the proposed method converges on the hyper-entropy range $[0.03, 0.50]$, which is more than twice as wide as the range $[0.04, 0.23]$ of the preceding moving method; it can therefore reflect vagueness and randomness of the concepts over a much wider range of uncertainty. Second, the overshoot of both the cart and the pendulum is reduced by roughly 10–40% with respect to T-S cloud modeling [14] in the convergent cases. Third, since the entropy is obtained without derivatives and integrals, the proposed method is about 2.8 times faster than the preceding moving method and about 1.9 times faster than T-S cloud modeling, which makes it suitable for real-time control loops—a property that recent inverted-pendulum studies with learning-based controllers [26, 27, 28] obtain only at a considerably higher computational cost.

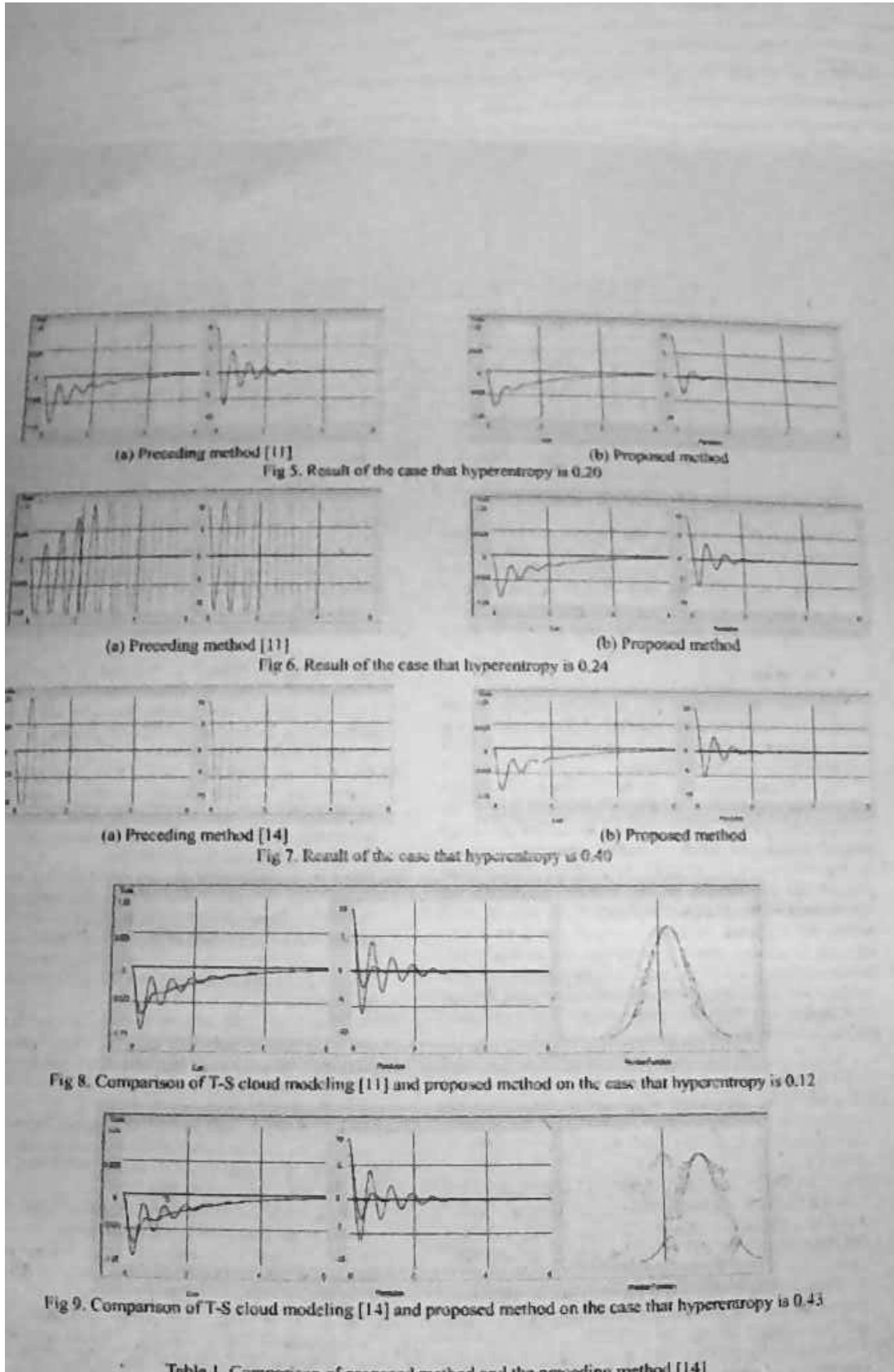


Figure 5: Results for the cases that the hyper entropy is 0.20, 0.24, and 0.40 (Figs. 5–7: (a) preceding method [11] / [14], (b) proposed method), and comparison of T–S cloud modeling [14] and the proposed method for the cases that the hyper entropy is 0.12 (Fig. 8) and 0.43 (Fig. 9), together with Table 1 of the draft.

6 Conclusions

This paper demonstrated that our study proposes a new membership cloud reasoning method that opens a novel research direction for approximate inference in nonlinear systems with uncertainty. To overcome the weaknesses of the preceding papers, we proposed a new approach of moving cloud reasoning to obtain the entropy of the membership cloud that does not use derivative and integral computation, based on the compensation fuzzy reasoning (moving-rate) principle proposed by Son-Il Kwak et al. in 2019 [12], and we corrected the errors contained in the condition expressions of the preceding formulation. Through the experimental research on stabilization of the inverted pendulum, the effectiveness of the proposed approach was shown: the proposed m -input single-output cloud reasoning approach by the moving principle overcomes the weakness of derivative and integral use in the preceding papers, reflects fuzziness and randomness as well as the three numerical characteristics of the cloud through cloudification and de-cloudification of the input information, satisfies the reductive property, converges in the wide hyper-entropy range $[0.03, 0.50]$, and reduces the average calculation time from 2058 ms (preceding moving method) and 1375 ms (T-S cloud modeling) to 742 ms.

In future work, we will extend the moving cloud reasoning method to interval and multi-output cloud models following the interval extension of the moving-rate principle [20], tune the cloud parameters automatically with metaheuristic scheduling such as improved shark smell optimization [22], combine the method with causal-entropic explainability analysis [21], and apply it to further nonlinear benchmarks and industrial plants.

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